MICHAELA BREZINOVA

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EDUCATION

- **UNIVERSITY OF CAMBRIDGE,** PhD in Computational Chemistry (Vendruscolo Lab) **Oct 2023 - Present**
    - Focus on advancing computational approaches to drug discovery, primarily for small molecules, with an emphasis on neurodegenerative disorders. See *Publications* section.
        - Using AI-driven active learning and screening approaches to develop pipelines for discovering binders to challenging targets, including Aβ42 disordered monomers, Aβ42 fibrils, TAF15 fibrils, and the design of subunit-selective ligands for GABAA receptors.
            - Identified the *first reported small-molecule binders* of Aβ42 fibril catalytic surfaces that inhibit secondary nucleation, with low-nanomolar affinities (GitHub link)
        - Developing an end-to-end ML pipeline to screen large chemical libraries for covalent inhibitors (KRAS G12C).
    - Other:
        - Performing high-throughput molecular dynamics simulations of nanobodies
        - Developing a web server for calculating critical concentration for phase separation (webserver link, GitHub link).
    - **Technical Skills and Tools:** *Python, TensorFlow, PyTorch, HPC (Slurm workload manager), Bash, AutoDock Vina, OpenEye Suite, RDKit, Open Babel, AlphaFold3, Boltz-2, Molecular Dynamics (GROMACS), Chimera(X)*

- **UNIVERSITY OF CAMBRIDGE,** MPhil in Computational Biology (**Distinction**) **Oct 2021 - Nov 2022**
    - **Relevant Coursework:** *Genomics I/II, Biological Imaging and Analysis, Deep Learning, Genome Sequence Analysis, Population Genetics Analysis*

- **UNIVERSITY OF EDINBURGH,** BSc Honours in Mathematics and Computer Science (**First Class**) **Sep 2016 - Jun 2020**
    - **Relevant Coursework:** *Honours Algebra, Differential Equations, Complex Variables, Probability, Bayesian Theory, Algorithms and Data Structures, Machine Learning Practical, Quantum Information, Computer Graphics*

WORK EXPERIENCE

- **UNIVERSITY OF CAMBRIDGE,** Research Assistant **(Cambridge, UK) May 2023 - Oct 2023**
    - Member of Team Wood as a part of the global Aligning Science Across Parkinson's initiative (ASAP).
        - Unsupervised learning analysis of DAB staining microscopy images of large alpha-synuclein aggregates from Parkinson's patients and developing a kinetic model of aggregate growth.
    - **Technical Skills and Tools:** *Python, TensorFlow, FIJI (ImageJ)*

- **ILLUMINA,** Bioinformatics Intern **(Cambridge, UK) Aug 2022 - Nov 2022**
    - Member of Genome Quality Group
        - Large-scale analysis of genomic data and development of a method for detecting variants in nucleotide sequences based on haplotype diversity. See *Patents* section.
    - **Technical Skills and Tools:** *Nextflow, Make, Bash, BCFTools, R*

- **GOOGLE,** Software Engineer at Google Health **(London, UK) Sep 2020 - Oct 2021**
    - Worked on Care Studio, clinical software to unify healthcare data, with a focus on desktop features and front-end latency optimisations
        - Delivering new features to the platform that optimise the data shown to clinicians (required full-stack work)
        - Collaborating with clinicians, product managers, as well as UX designers to make sure the features work exactly as intended and improve the customer experience
        - Designing (under supervision), implementing and delivering the first version of a *critical latency feature required to meet the release latency requirements*
    - **Technical Skills and Tools:** *Soy (Closure templates), TypeScript, Java, C++*

- **CERN,** Remote Summer Collaboration **(Remote) Jun 2020 - Aug 2020**
    - Worked on a profile customisation for ALICE (A Large Ion Collider Experiment) O2 InfoLogger (https://cds.cern.ch/record/2728718)
        - Planning and delivering a new profile customisation for ALICE O2 InfoLogger, taking *software robustness and durability (10+ years)* into account (once the experiment resumes, the program has to be functioning for 10+ years)
    - **Technical Skills and Tools:** *HyperScript, JavaScript, NodeJS, SQLite*

- **GOOGLE,** SWE Intern at Google Travel NBU (Next Billion Users) **(Zurich, Switzerland) Jul 2019 - Sep 2019**
    - Worked on a new flexible dates feature for a train booking platform integrated in Google Pay for the Indian market (India Rail Travel App).
    - **Technical Skills and Tools:** *Soy (Closure templates), TypeScript, Java*

- **GOOGLE,** STEP Intern at Google Account **(Munich, Germany) Jul 2018 - Sep 2018**
  - Worked on the redesign of the Takeout page (takeout.google.com).
  - The new design is currently in production.
  - **Technical Skills and Tools:** *Soy (Closure templates), TypeScript, Java*

PATENTS

- Andrews, D.J., **Brezinova, M.**, 2024. Detecting variants in nucleotide sequences based on haplotype diversity (https://patents.google.com/patent/WO2025090883A1/)

PUBLICATIONS

- **Brezinova, M.**, Brotzakis, Z.F., Horne, R.I., Roy Chowdhury, V., Gregory, R.C., Bian, Y., González Díaz, A., Gentile, F. and Vendruscolo, M., 2025. Identification of potent high-affinity secondary nucleation inhibitors of Aβ42 aggregation from an ultra-large chemical library using deep docking. *Molecular Systems Biology*, pp.1-19. https://doi.org/10.1038/s44320-025-00159-5 (GitHub link)

- **Brezinova, M.**, Fuxreiter, M. and Vendruscolo, M., 2025. DropFit: Determination of the Critical Concentration for Protein Liquid-Liquid Phase Separation. *Journal of Molecular Biology*, p.169294. https://doi.org/10.1016/j.jmb.2025.169294 (webserver link, GitHub link)

- Amico, T., Dada, S., Lazzari, A., **Brezinova, M.**, Trovato, A., Vendruscolo, M., Fuxreiter, M., & Maritan, A. (2024) 'A scale-invariant log-normal droplet size distribution below the critical concentration for protein phase separation', *eLife*, 13, RP94214. https://doi.org/10.7554/eLife.94214.2

TALKS

- **Talk at In Silico #001 event series (https://luma.com/dw9ijcrm) (London, UK) Jul 2025**
- **Short Talk at AI in Drug Discovery and IRB Barcelona Biomedicine Conference (Barcelona, Spain) Apr 2025**
- **Talk at Bio2Brain Network's (https://bio2brain.eu/) AI Workshop (Online) Jun 2024**

SKILLS

- **Programming Languages:** ***Python*** *(experienced),* ***Bash*** *(experienced),* ***TypeScript*** *(experienced),* ***Soy (Closure Templates)*** *(experienced),* ***HTML*** *(experienced),* ***CSS*** *(experienced),* ***Java*** *(skilful),* ***R*** *(skilful),* ***C++*** *(skilful),* ***Haskell*** *(familiar)*
- **Cheminformatics:** ***AutoDock Vina*** *(experienced),* ***OpenEye Suite*** *(experienced),* ***RDKit*** *(experienced),* ***Open Babel (Obabel)*** *(experienced),* ***AlphaFold3*** *(skilful)**,* ***Boltz-2*** *(skilful)**,* ***Molecular Dynamics(GROMACS)*** *(familiar),* ***Chimera(X)*** *(skilful)*
- **Machine Learning:** ***TensorFlow*** *(skilful),* ***PyTorch*** *(familiar)*
- **Computational Tools and Frameworks:** ***Slurm*** *(experienced),* ***Nextflow*** *(skilful),* ***Make*** *(skilful)*
- **Spoken languages:** ***Slovak*** *(native proficiency),* ***Czech*** *(native proficiency),* ***English*** *(full professional proficiency),* ***German*** *(limited working proficiency),* ***Mandarin Chinese*** *(elementary proficiency)*

VOLUNTEERING AND EXTRACURRICULAR - ACADEMIC

- **Lab Demonstrator at the University of Cambridge (Cambridge, UK) Jan 2024 - Feb 2025**
  - Lab demonstrator for computational practicals of Chemistry Part IB course. The main focus is the use of Python, Google Colab, Avogadro and Orca

- **Code First: Girls, Introduction to Web Development course instructor (Edinburgh, UK) Jan 2020 - Mar 2020**
  - Teaching female students about the front-end development: HTML, CSS, Bootstrap, Git and GitHub

- **Director of Media for the Women in Tech Conference in Edinburgh (Edinburgh, UK) May 2017 - Jun 2018**
  - Organising Women in Tech Conference in Edinburgh, focusing on media and graphic design

- **MathPALS Student Leader at the University of Edinburgh (Edinburgh, UK) Sep 2017 - Dec 2017**
  - Facilitator at Mathematics Peer Assisted Learning program, holding study groups for first-year Mathematics students

VOLUNTEERING AND EXTRACURRICULAR - OTHER

- **Secretary at the Cambridge University Massage Society Jun 2024 - Present**

COMPETITION ACHIEVEMENTS

- 3rd place at International Team High School Internet Mathematical Olympiad
- 3rd place at National Scientific Research Paper's Competition for high school students in Mathematics and Physics category with paper on topic *Mathematics and Music*